

Title: Modeling the structure of DNA: Latent Distance Models for sparse single-cell data

Supervisor: Oscar Lauritz Møberg (osmo@dtu.dk), Section for Cognitive Systems, DTU Compute

Motivation

Single-cell ATAC-seq (scATAC-seq) is a measurement technique that tells us, for each individual cell in a human tissue sample, “*which pieces of DNA are physically open and accessible to the cell’s molecular machinery?*”. Accessibility matters because it is closely linked to the gene regulation, cells often control which genes are active by opening or closing specific regulatory regions, so scATAC-seq provides information that complements what you get from gene sequencing (RNA) measurements giving us a clearer image into why a cell behaves as it does, and what is it genetically capable of doing. (Buenrostro et al., 2015)

From a data perspective, scATAC-seq is challenging because it is high-dimensional and extremely sparse. We define a large set of candidate genomic regions (“peaks”), often on the order of 10^5 - 10^6 . For each cell, we only observe evidence for accessibility at a tiny fraction of these peaks, so the resulting cell-by-peak matrix is dominated by zeros. Importantly, many zeros are not true negatives, they arise because each cell is only sampled shallowly, so “*not observed*” can mean “*missed observation*,” not necessarily inaccessible. The combination of rare observations + measurement noise makes modeling nontrivial and motivates probabilistic approaches that explicitly account for uncertainty.

Because of this sparsity, many standard workflows rely on heavy aggregation and dimensionality reduction, and there’s a growing ecosystem of probabilistic/deep generative approaches for scATAC-seq representation learning (e.g., topic models, VAEs). (Ashuach et al., 2022; Bravo González-Blas et al., 2019; Xiong et al., 2019).

In this project, we’ll explore a complementary idea, a **Latent Distance Model (LDM)** (Hoff et al., 2002) to jointly embed *cells* and *peaks* in a shared metrics space where distances directly reflect the probability of genome accessibility in a cell. The LDM also provides a clean handle for injecting biology-informed side information, like genomic proximity or peak–peak connectivity (co-accessibility). (Pliner et al., 2018)

Project description

You will apply a **Latent Distance Model (LDM)** to scATAC-seq data, with the cell-peak matrix viewed as a bipartite graph. The goal is to learn a low-dimensional embedding where distances explain which edges are observed in the graph. You will evaluate the approach using predictive performance on held-out data and validate the biological coherence of the learned embeddings, e.g. by recovering known cell types. If time permits you will try to extend the baseline model by incorporating side-information and study how it affects the learned latent space and relevant evaluations.

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Title: Graph-Based Latent Distance Models for Data Visualization

Supervisor: Oscar Lauritz Møberg (osmo@dtu.dk), Section for Cognitive Systems, DTU Compute

Motivation

Dimensionality reduction is useful whenever data live in more than 2–3 dimensions, this is especially true when trying to visualize complex systems and their embeddings and feature spaces as in representation-based models for network, language, recommender systems, sensor data, and many other domains. Common tools such as t-SNE, UMAP, and PaCMAP produce effective low-dimensional layouts by optimizing objectives that emphasize neighborhood structure with different trade-offs between local and more global relationships. (van der Maaten & Hinton, 2008; McInnes et al., 2018; Wang et al., 2021).

This project explores a complementary perspective: treat dimensionality reduction as learning a low-dimensional latent space that explains a sparse graph of relationships between points, using a **Latent Distance Model (LDM)**. In an LDM, relationships are more likely (or stronger) when latent points are closer, which provides a likelihood-based framework for fitting and evaluation. (Hoff et al., 2002)

Project description

In this project, we will develop a dimensionality reduction pipeline rooted in the LDM, likely implemented in PyTorch. The core objective is to learn interpretable low-dimensional embeddings (2D or 3D) from generic high-dimensional data (or simply higher dimensional embeddings) by modeling sparse pairwise relations. This work will require exploring different strategies for constructing input graphs from raw data, balancing local neighborhood structure with global connections using principles of attraction and repulsion. Finally, the utility of the approach will be assessed by benchmarking its ability to preserve data structure against established dimensionality reduction baselines. (van der Maaten & Hinton, 2008; McInnes et al., 2018; Wang et al., 2021).

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Title: Decoding auditory attention in speech perception from EEG data
Supervisor: Tobias Andersen (toban@dtu.dk), Section for Cognitive Systems, DTU Compute

Motivation

In neuroscience, we are often interested in measuring the brain's response to a stimulus. Measurements of brain activity are typically very noisy and reflect also ongoing neural activity that is not related to processing the stimulus. Trial-based paradigms overcome this problem by presenting the stimulus across many trials. When averaging the measurements across trials, the random irrelevant activity is removed leaving mostly the neural response to the stimulus. The problem with trial-based paradigms is that they do not reflect the complexity of human behaviour.

In this project we will study neural responses measured with EEG to speech stimuli. In a trial-based paradigm, we would have to present the same speech stimulus across many trials. This does not reflect the complexity of speech communication. Therefore, we will use novel encoding models for relating brain activity to speech stimuli. These methods model the temporal response function (TRF) of EEG with respect to speech stimuli. This encoding model allows us to model how the neural responses encode full speech sequences. One application of this approach is auditory attention decoding (AAD). When two speakers speak simultaneously, we can direct our attention to either speaker. Studies have shown that the attended speech signal is encoded more strongly in the EEG signal than the unattended speech signal. In this way we can decoded the attentional state of the observer (Fuglsang et al., 2017).

Project description

In this project we will analyse EEG data from speech perception experiments by fitting the TRF of EEG with respect to the speech signal. Then we will try to reproduce previous from a study on AAD. We can explore different methods for estimating the TRF including linear and neural network methods (Brodbeck et al., 2023; Nguyen et al., 2025). We can also explore the effect of being able to see the talker while listening on the TRF (Man et al., 2025).

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Day-trading as a multilevel decision problem

Tue Herlau, tuhe@dtu.dk

TL;DR: This project will deal with automated day trading, i.e., buying and selling stocks within a short time frame to maximize profit while being risk averse. The challenge is to build a trading algorithm and carefully evaluate it.

Who is this for: This project will require downloading and working with stock market data within a python framework that allows cross-validation of the methods under market conditions; this task will need to be carried out in python before you can implement the first algorithm. An ideal group will have a very hands-on approach to this challenge and be able to distribute it among several team members so that the project can get an initial implementation off the ground early in the project period.

The Challenges: We can think of day trading as a decision problem in which our trading method must decide which of N stocks to potentially buy/sell at a given time T , based on their historical price of a stock i , $y_i(t) \in \mathbb{R}$, and side information $x_i(t) \in \mathbb{R}^d$, $t = 1, \dots, T-1$. The accumulated difference in buy/sell price, $y_i(t+\Delta) - y_i(t)$, after transaction fees, then defines the profit over a time period Δ . The relevant time resolution, t , is 1-5 minutes. From a machine-learning perspective, day trading provides an interesting combination of challenges

- The problem is non-stationary (i.e., time-dependent) due to outside forces such as macroeconomics and market sentiment,
- The algorithms must be online and learn/act in response to streaming data,
- Day trading is notoriously difficult to cross-validate correctly; (the more mistakes the better are the results).

The decision problem is furthermore unusual in that it is highly asymmetrical: in nearly all cases, the best action is to do nothing and thereby not lose money, perhaps for days at a time, which makes the derived machine-learning problem highly unbalanced.

Approach: A popular strategy to circumvent this problem is to apply a staged trading approach, where the first stage apply rule-based *filters* and the last stage makes the actual (machine-learning-based or often expert human) *enter-trade* decision [LdP18]:

1. Filter out few interesting stocks, for example based on the previous day's behavior or the trading volume,
2. Filter the time points based on simple rules such as whether the stock seems to be in an upward trend,
3. Only make machine-learning based trading decisions for stocks i and times t where conditions (1) and (2) are met.

In the project, you will try to formalize the *filter+enter* decisions as a joint optimization problem and thereby obtain a single optimization objective which corresponds to profit. The hope is that the joint optimization will mean that relatively simple decision models are sufficient, and that they by their simplicity can avoid overfitting. To sketch the objective:

$$\text{Profit}(\theta) = \sum_{i=1}^N \sum_{t=1}^T P_{\text{filter}}(i, t \mid \dots, \theta) P_{\text{enter-trade}}(\text{Enter trade at cost } y_i(t) \mid t, i, \dots, \theta) R(y_{i,t}, y_{i,t'})$$

Here, $P_{\text{filter}}(i, t \mid \dots, \theta)$ is the probability that we consider a stock (filtering), $P_{\text{enter-trade}}(\text{Enter trade at cost } y_i(t) \mid t, i, \dots, \theta)$ is the probability that we enter a trade at time t (and decide to exit at time $t + \Delta$), and $R(y_{i,t}, y_{i,t+\Delta})$ computes the cost-aware profit. To implement the decision rule itself, we will use a triple-barrier objective¹. An important aspect of the project is to carefully implement and test the strategy in a realistic setting. To this end, we will use the **vectorbt** Python framework.

Objectives

- Create a database of high-resolution trading data and implement a realistic cross-validation framework for a trading algorithm using a framework such as **vectorbt**.
- Design and implement a basic staged algorithm and evaluate its historical performance compared to macro trends.
- Formulate a joint optimization objective and compare this approach with the baseline

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¹see [GWL25] for a public writeup; our approach will be simplified

A Comparative Analysis of Classification Approaches for Discrete Data

Supervisor: Kazu Ghalamkari (kazfu@dtu.dk), Mads Emil Kofoed Rehof (mekre@dtu.dk), Morten Mørup (mmor@dtu.dk), Section for Cognitive Systems, DTU Compute

Motivation

Discrete inputs are very commonly occurring in many classification problems. I.e., when considering for instance questionnaire data or general categorical datasets. In this project, various classification approaches tailored for discrete inputs will be implemented and their strength and weaknesses systematically investigated on a variety of classification benchmark datasets to understand the merits and limitations of tensor network and many body approximation based approaches.

Project description

Specifically, the project will develop a PyTorch framework for classification using discrete inputs considering standard neural network based modeling approaches, as well as tensor network based modeling procedures [1], and many-body-approximation (MBA) methodologies [2]. Notably, the MBA forming convex optimization problems. Strength and limitations of the considered approaches will be systematically investigated and their performance assessed considering a variety of discrete classification benchmarks [3].

References

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Group equivariant deep learning

Introduction

Deep learning models often struggle with data that exhibits symmetries, such as rotations, reflections, or translations. Group equivariant deep learning addresses this by designing architectures that respect these symmetries, leading to more efficient and interpretable models. By incorporating group theory into neural network design, these methods improve generalization, reduce data requirements, and enhance performance in tasks such as computer vision, physics simulations, and molecular modeling.

This project will introduce students to the foundational concepts of group equivariance in deep learning, explore its applications, and culminate in implementing a group-equivariant model for a specific dataset or task. The project is ideal for students with a basic understanding of deep learning who wish to deepen their knowledge of advanced model design principles.

Learning objectives

1. **Understand Group Theory Basics:** Learn the fundamental concepts of group theory, including group actions, representations, equivariance, and invariance.
2. **Explore Equivariance in Neural Networks:** Study how group equivariance is implemented in neural network layers, such as convolutional layers.
3. **Analyze Applications of Equivariant Models:** Investigate real-world use cases where group equivariant methods offer significant advantages.
4. **Implement and Evaluate Models:** Build and train a group-equivariant deep learning model using a library such as PyTorch.
5. **Compare Performance:** Evaluate the model's performance against traditional architectures and analyze the benefits of incorporating equivariance.

References

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Supervisor

Mikkel N. Schmidt (mnsc@dtu.dk)

Title: On-Device Training on Arduino from Streaming (Peripheral Nerve) Signals

Supervisor: Petr Taborsky (*ptab@dtu.dk*), Morten Mørup, Cognitive Systems, DTU Compute

Motivation: Modern embedded systems increasingly run machine learning locally (TinyML), enabling low-latency, privacy-preserving adaptation without a cloud connection. This project goes one step further than typical TinyML: *training / fine-tuning on-device (in C++)*. A motivating biomedical direction is peripheral nerve recovery, where electrical stimulation and electroactive biomaterials are studied to support regeneration and functional recovery [Song et al., 2025]. However, even outside medicine, on-device learning matters whenever signals drift over time (sensor placement, temperature, fatigue, device-to-device variation) and models must adapt *in the field*.

(Dictionary of terms: **Peripheral nerve** carries motor/sensory signals between spinal cord and body (e.g., *sciatic nerve* in the leg). **ENG** (electroneurography) stands for electrical recordings from a nerve (often via *cuff electrodes*. Signals are low SNR and multi-channel. **CAP/CNAP** (Compound action potential), aggregate nerve response from many fibers, often used for classification/decoding tasks.)

Project description: Groups work with Arduino Nano 33 BLE Sense Rev2 board that will be provided for this project duration and implement an end-to-end *on-device learning* pipeline. Students may (A) propose their own dataset/task/model (subject to feasibility on the Arduino board), or (B) use the provided default workflow and dataset.

In case you opt for default peripheral nerve recovery task: A real deployment is to be simulated by *replaying a real ENG dataset as a live stream* to the Arduino over serial. The default dataset is an open multi-channel sciatic-nerve cuff recording dataset (MATLAB) released as replication data [Zariffa, 2023, Koh et al., 2019], also used in later deep-learning benchmarking work [Gill and Zariffa, 2024].

Suggested tasks (milestones):

M1. Choose a problem. Define your own task/dataset. *Optionally:* Select one of 2-3 predefined labels (e.g., stimulus class/pathway class/state class) from the default ENG dataset, that has been used to train ML models successfully, [Koh et al., 2019].

M2. Design a tiny model. Pick a compact architecture/model (e.g., linear classifier, small MLP, tiny 1D CNN, Random Forests, etc.) and decide what to simplify for robust C++ implementation (activations, normalization, etc.). Document and justify trade-offs.

M3. Implement training on-device. Note, that while Arduino supports micro python, the Arduino C++ is strongly recommended. Implement your learning algorithm. If you choose neural network model, implement forward pass, loss, backprop gradients, and an optimizer (SGD or Adam-lite). Focus on numerical stability, memory limits and test the scaling of the model vs. performance. Present the results.

M4. Stream & learn online. Consume the live replay stream on Arduino (fixed-rate windows), update the model online, and log performance over time (learning curve).

M5. Demonstrate adaptation. Show improvement during streaming and/or recovery after an induced shift (e.g., changed noise level / channel dropout / distribution split). Report accuracy, update time, and memory footprint.

Deliverables: (i) Arduino C++ training code (including your implementation of backprop or other optimization method of your choice), (ii) the streaming/replay tooling used, (iii) a technical report (problem, model, training details, experiments), (iv) final demo (live stream → on-device learning).

References

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Title: Quadratic Neurons vs. ReLU: How Much Nonlinearity Is Enough?

Supervisor: Petr Taborsky (*ptab@dtu.dk*), Morten Mørup, Cognitive Systems, DTU Compute

Motivation: Modern high-performing deep networks often rely on expensive pre-training and increasingly complex architectures. QuadraNet V2 proposes a sustainable alternative: reuse an existing (pretrained DNN) backbone and adapt it by finetuning only added quadratic terms (a “Quadratic Adapter”), see Figure 1 from [Xu et al., 2024], reaching state of the art accuracies for fraction of additional GPU hours. At the same time, quadratic transformations are a principled way to inject feature interactions with modest overhead, and connect naturally to polynomial networks and classical geometry (quadratic forms). [Chen et al., 2025]. The resulting QuadraNet V2 network is a hybrid, that comprises two kind of non-linearities, quadratic neuron terms and standard (e.g., GELU/ReLU) ones.

This project explores a fundamental question with both theoretical and practical implications: *To what extent are standard pointwise nonlinearities and quadratic neurons complementary versus redundant?* If a quadratic-adapted model can be converted into a “pure” polynomial (no pointwise nonlinearities), we gain a cleaner mathematical object for analysis and potential interpretability, but we may lose accuracy or robustness, [Fan et al., 2023].

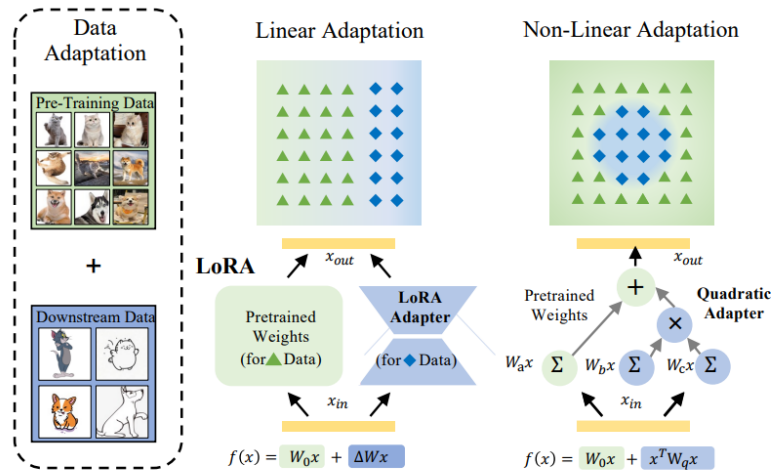


Figure 1: Linear vs. Nonlinear Data Adaptation (Fig. 1 in the [Xu et al., 2024]).

Project description: The project combines **Theory** and **Practice**. Students may choose one as the main focus (and keep the other lightweight). The outcome is an experimental study and a report combining theory with plots and targeted ablations.

A. What is lost in translation? Build a *more polynomial-like* variant of a quadratic-adapted network by reducing/removing standard nonlinearities, and discuss what changes in expressivity and inductive bias. Experimentally compare accuracy/generalization of the original vs. the simplified variant (in-distribution + at least one mild shift/corruption setting).

B. Quadratic adaptation vs. training from scratch On a small backbone, train (i) with frozen base + trainable quadratic terms and (ii) fully from scratch. Compare performance and efficiency (e.g., accuracy vs. training budget) and reflect on when quadratic adaptation helps or not and why.

C. Geometry and interpretation of quadratic terms Can learned quadratic terms be geometrically interpreted (e.g. a distance/metric on latent representations)? Analyze the learned quadratic parameters at a high level (e.g., magnitude, spectrum/conditioning, dominant directions, layer-wise patterns). Do one simple intervention study (ablation/perturbation) to probe how quadratic terms relate to generalization and robustness.

Expected deliverables: (i) reproducible code (training + evaluation + ablations), (ii) a report with clear experimental protocol and plots, and (iii) a presentation.

References

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Scalable Tensor Decomposition beyond Euclidean Formulation

Supervisor: Kazu Ghalamkari✉ and Morten Mørup✉

Background: Tensor (multidimensional array) is a fundamental data structure in machine learning. We can extract patterns, relevant information, and features underlying data by decomposing tensor-formatted data into a sum of product form as

$$\mathcal{T}_{ijk} \approx \mathcal{P}_{ijk} = \sum_{r=1}^R A_{ir} B_{jr} C_{kr}.$$

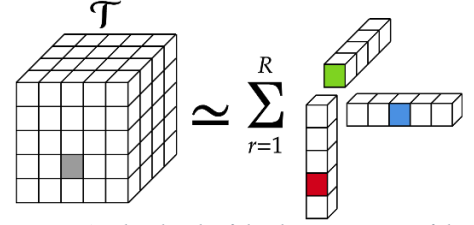


Figure 1: The sketch of the decomposition of the tensor T . Each vector in right side corresponds to $A[:,r]$, $B[:,r]$, and $C[:,r]$.

However, naïve implementation of this task, minimizing the Frobenius error $\|\mathcal{T} - \mathcal{P}\|_F$ inherits ill-posedness and NP-hardness even in the simplest rank-1 setting ($R=1$). Nowadays, to overcome these challenges, our teams have introduced two alternative suitable reformulations: (i) extending EM (Expectation-Maximization) based tensor decomposition, minimizing the Amari-divergence $D_\alpha(\mathcal{T}, \mathcal{P})$ with closed form updates [1], and (ii) *deformed decomposition*, as

$$\min D_\chi(\mathcal{T}, \mathcal{P}) \text{ such that } \mathcal{P}_{ijk} = \sum_{r=1}^R A_{ir} \otimes_\chi B_{jr} \otimes_\chi C_{kr},$$

for any given increasing monotonic function $\chi: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ where the corresponding generalized product $\otimes_\chi$ and χ -divergence $D_\chi(\mathcal{T}, \mathcal{P})$ relaxed the optimization into an iterative convex optimization problem.

Project purpose: While the above decomposition beyond Euclidean formulation provides implicit regularization and robustness against noise, our current implementation remains limited in scalability. We offer research opportunities to scale these decompositions to large datasets on a practical problem setup using a stochastic-based method, and to investigate the model's behavior in an overparameterized regime. We further explore the application of these decompositions for modeling data and performing density estimation.

Requirements: The project will in particular leverage skills in (i) basic Python and PyTorch programming, and (ii) foundational knowledge of linear algebra.

References:

[1] Kazu Ghalamkari, Jesper Løve Hinrich, Morten Mørup, E²M: Double Bounded α -Divergence Optimization for Tensor-based Discrete Density Estimation, Preprint, [arXiv:2405.18220](https://arxiv.org/abs/2405.18220)

Tensor Networks for Reinforcement Learning.

Supervisor: Mads Emil Kofoed Rehof (mekre@dtu.dk)

Reinforcement learning (RL) traditionally relies on representing and learning value functions, Q-functions, policies, or even full transition models[1], [2]. A common way to do this is via neural networks and deep RL[3]. While this approach has been extremely successful, neural networks can be large, difficult to train, and may struggle to exploit latent structure in environments with many interacting variables.

Tensor networks (TN), including Tensor-Train (TT/MPS), Tucker, and other structured factorizations, offer an alternative way to represent high-dimensional functions using compact, interpretable components[4]. Recent work has shown that RL problems and Markov decision processes can be formulated naturally using tensor-network representations of trajectories and transitions, suggesting that TNs may provide new tools for learning or compression in RL[5][6].

Other results show that low-rank tensor decompositions can reveal latent structure in state–action dynamics, with potential applications in approximate value function learning or model-based RL[7]

Project Description

The project investigates how tensor networks can be used within reinforcement learning. The exact direction is flexible and can be adapted to the students' interests. Possible components include (but are not limited to):

- Using tensor networks as function approximators, replacing or complementing neural networks for value functions, Q-functions, or policies. This may involve experimenting with Tensor-Train (TT/MPS) or Tucker representations.
- Exploring tensor-network representations of RL trajectories, inspired by tensor-based formulations of finite-horizon MDPs
- Applying low-rank tensor decompositions to state–action spaces to uncover latent structure or compress high-dimensional RL models.
- Implementing RL agents in simulated environments (e.g., OpenAI Gym) using tensor-network components.
- Comparing performance of tensor-network-based agents with standard deep RL agents, examining aspects such as sample efficiency, model size, and stability.
- Studying representational benefits: when do tensor networks offer advantages, and what kinds of environments or state structures make them particularly suitable?

The project is exploratory: students may focus on algorithmic development, experiments, theoretical understanding, or a combination of these.

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Levering EEG Models for Better Neurological Care in Kenya

BrainCapture Collaboration Project Proposal

Supervisor: [William Lehn-Schiøler](#), Industrial PhD at BrainCapture and DTU Health Tech.

Contact: wthle@dtu.dk, wls@braincapture.dk

Background

The integration of BrainCapture's portable EEG device with advanced Artificial Intelligence (AI) addresses the global need for accessible neurological diagnostics, shifting the paradigm from centralized expert interpretation to automated analysis at the point of care. This ambitious technical challenge requires cutting-edge deep learning, including **Transformer models**, to effectively process complex EEG time-series data, capturing crucial long-range temporal dependencies that are key to identifying biomarkers for conditions like epilepsy. Crucially, the deployment must incorporate **Explainable AI (XAI)** techniques to ensure clinical trust by visualizing *why* a diagnosis was made, thereby bridging the gap between sophisticated algorithms and medical practice and offering high-impact project opportunities in fields like artifact removal, biomarker feature extraction, and real-time model optimization for low-resource environments.

Project Proposals

- 1. Implement and benchmark powerful EEG models:** State-of-the-art for modelling EEG signals is constantly being challenged with new model paradigms being presented on a monthly basis. In this project, you evaluate the newest models on real production data.
- 2. Explore advanced signal processing techniques for feature extraction and explainability of EEG models:** EEG signals are difficult to interpret by non-experts and a substantial potential can be found in combining traditional feature extraction with advanced modelling. Interpretable biomarkers can, moreover, assist in explaining model decisions.
- 3. Mapping EEG to Text Embeddings:** Most EEG recordings are linked to a clinical report. We can use textual descriptions of EEG to learn better representations of brain signals by applying techniques known from image diffusion models.

Optimal Transport for Generative Modelling in JAX

Supervisor: Bjørn Sand Jensen, bjje@dtu.dk

Optimal transport (OT) enables domain adaptation and unpaired generative modeling, making it possible to translate data between different domains in a principled manner without requiring aligned training data. Originating in probability theory, it is widely used in machine learning to measure distances between distributions, define loss functions for generative models, and learn transformations between distributions from unpaired observations. In this project, we focus on this latter application, which allows, for example, translating images of human faces into images of animals without ever observing paired examples.

The objective of this project is to study a basic formulation of optimal transport for small-scale, low-dimensional problems [2], potentially extend it to amortized OT variations using neural networks [3,4], implement selected methods in JAX, and apply them to an interesting application domain such as image-to-image translation using low-dimensional representations learned via autoencoding.

Potential Project Elements: The exact scope will be determined together with the assigned students. Possible elements include:

- study the regularised optimal transport problem and implement basic Sinkhorn algorithms [2, Ch 4] (and potentially amortised variants [2,3]) in Python using JAX.
- study and implement machine learning methods for finding low-dimensional representations of high-dimensional objects (e.g. images) via autoencoding [1]
- apply discretised/approximate Sinkhorn-style algorithms to toy data and a relevant image dataset using an autoencoded (possibly discrete) latent space.
- evaluate performance using relevant metrics, including reconstruction loss, FID and various statistical approaches.
- if time permits, studying and comparing classical Sinkhorn variants with recent bridge formulations [4,5] that aim to scale OT, but results in certain inference time issues.

You should choose this project if you:

- would like to study representation learning (here: autoencoding) using deep neural networks.
- appreciates probability and information theory as means of formulating principled yet simple algorithms in generative machine learning.
- would like to learn about amortisation of algorithms in machine learning using function approximators such as neural networks.
- enjoy implementing numerical algorithms in Python using NumPy, PyTorch, or JAX (you would likely need to learn JAX as part of the project).

Literature

1. Buchanan et al., Learning Deep Representations of Data Distributions, 2025. <https://ma-lab-berkeley.github.io/deep-representation-learning-book/>
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3. Aude Genevay et al. *Large-Scale Optimal Transport and Mapping Estimation*. ICLR, 2018.
4. De Bortoli et al, Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling, 2021. <https://arxiv.org/abs/2106.01357>
5. Liu et al, I2SB: Image-to-Image Schrödinger Bridge, 2023, <https://arxiv.org/abs/2302.05872>

Note: some of these references are unnecessarily math-heavy, using notation from measure theory (which you have most likely not heard about yet), but in our case, most if not all of the key algorithms can be boiled down to probability measures, meaning that we can deal with random variables as you have heard about in e.g. 02450 leading to relatively simple algorithms.

Titel: Approximate KV-Cache Compression for Fast Inference in Transformer-Based Models

Supervisors: Jesper Løve Hinrich (jehi@dtu.dk) and Anders Vestergaard Nørskov (aveno@dtu.dk)

Motivation: Fast inference is an important challenge for modern transformer-based models, especially large language models that operate on long input texts. When generating text token by token, models must repeatedly attend to long contexts, which makes inference slow and resource intensive. Improving inference efficiency is therefore important both for practical deployment and for reducing computational cost. To reduce repeated computation during text generation, modern transformer models use key and value (KV) caches that store information from previous tokens. For long inputs, the growing KV-cache must be stored and accessed at every generation step, so approximating or compressing it reduces both memory use and computation, making it a promising avenue for improving the efficiency of LLM inference.

Data/Model: This project will focus on long-range datasets such as LongBench, and use openly available local LLM models (e.g., llama-3.1-8b-instruct, phi-3.5-mini-instruct) or any other transformer model of your choice.

Project Description: This project explores how inference in transformer-based models can be made faster by approximating or compressing the key-value (KV) cache. The project investigates whether the information stored in the KV-cache can be represented in a more compact or approximate form while still producing high-quality model outputs.

The work will initially look at approximation strategies such as low-rank representations, clustering, or sparsity-inducing transformations of the KV-cache and related operations to reduce the cache and ease next-token computation. Two natural avenues are to compress the KV-cache for a fixed long context such as a system prompt, and/or to explore online methods that keep the cache compressed as new tokens are added during generation, with evaluation focusing on the trade-off between computational efficiency and output quality.

Title: Low rank plus sparse tensor decomposition for characterizing fluorophores in excitation- emission spectroscopy

Supervisors: Jesper Løve Hinrich (jehi@dtu.dk) and Morten Mørup (mmor@dtu.dk)

Motivation: Excitation-emission spectroscopy is a widely used technique for characterization of molecules and chemicals in a wide range of real-life applications, such as uncovering chemical reactions, environmental monitoring, pharmaceutical production and many more. It essentially works by lighting a sample at a specific wavelength of light (excitation), turning of the light and seeing the light given off by the same at specific wavelengths (emission spectrum), this is repeated for each sample. When multiple emission frequencies are used, each measured sample an excitation-emission matrix (EEM) and these are then collected in a third order tensor (three-way data array).

The data ideally follows the so-called Beer-Lambert law, that observed signal for each fluorophore at each emission and excitation wavelength is a product of the three. For multiple fluorophores this is a mixture problem and can be addressed through PARAFAC/Canonical polyadic decomposition. However, structured noise sources – in particular Rayleigh and Raman scattering - can contaminate the data and results in violation of the model assumptions.

Data/Model: This project will focus on EEM data from a set of beer samples which are diluted with water at different ratios (from 1:0 to 1:511) and as dilution increases the fluorophore signal decreases and the structured noise increases.

Project Description: The project will focus on tensor/multi-way decomposition – in particular PARAFAC model – but under the framework of robust tensor decomposition to automatically account for the structured noise. Robust tensor decomposition seeks to describe the data by a low rank contribution (the PARAFAC model) and an additive sparse array (structured noise).

Indices with Rayleigh scattering can be treated as missing – if they do not overlap with the fluorophore signals – or through a rank 1 matricized component. In contrast, Raman scattering often lies on top of the signal of interest and cannot easily be removed and treating the indices as missing would remove valuable information.

The goal is thus to separate the underlying fluorophores by jointly learning the model structure and the structured noise. The methods will be tested on EEM data, but will solve address a general problem in fluorescence spectroscopy.